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Near-Field Photometry: A New Approach

by

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Abstract

*This paper presents a new approach to near-field photometry and describes a novel near-field goniophotometer and illuminance calculation method. The approach is based on **helios**, which is a rationalization of the luminance concept for volume sources. The goniophotometer measures the three-dimensional vector field of light surrounding a luminaire rather than any of its intrinsic properties. The calculation method can predict the illuminance at any point on any surface anywhere within the 3-D space surrounding the luminaire. It does not require any knowledge of the geometry of the luminaire, nor does it require that the distance of the luminaire from the surface being illuminated be known.*

Introduction

In "A Study of Near-Field Indirect Lighting Calculations," Mistrick and English [1990] investigated the ability of lighting calculation programs to model ceiling luminances near an indirect luminaire. They concluded that:

"The current calculation practice of utilizing far-field photometric data in near-field applications does not accurately model ceiling luminances near an indirect luminaire ... to advance the state-of-the-art in lighting calculations, we must investigate alternative measurement and calculation procedures that would permit us to accurately model the near-field environment."

There have been several attempts to develop such procedures. Stannard and Brass [1990] developed Application Distance Photometry, which models the luminaire as an equivalent point source, while Franck [1950] and Lautzenheiser et al. [1984] modeled the luminaire as an array of point sources. Verbeck and Greenberg [1984] and Ngai [1987] examined the theoretical aspects of modeling the luminaire as an array of point sources, but did not offer any measurement procedures.

Application distance photometry offers an approximation of the near-field environment for specific planes of measurement. Bradford and Stannard [1992] and Ngai et al. [1992] demonstrated that the illuminance of planes parallel to and between those measured can be reasonably interpolated from the photometric data. However, this data cannot be extrapolated to planes that are tilted with respect to those measured, or to any plane that intersects the luminaire volume (Moon and Spencer [1981]).

Modeling the luminaire as an array of point sources requires that the luminous intensity distribution and spatial location of each source be known. The illuminance of a surface in the near-field environment can then be determined by using the inverse square law to calculate the luminous flux contribution of each source.

Ngai [1987] demonstrated that the luminous intensity distribution of each infinitesimal point on the surface of a luminaire can be calculated if the luminance distribution function of each point is known. Unfortunately, it is not possible to accurately measure these functions in that we are limited to measuring the luminance of finite areas. If the surface is convex or concave, the area will partially occlude itself at oblique angles, or be occluded by other parts of the luminaire.

Clearly, modeling a luminaire as a point source or array of point sources does not offer a practical technique for accurately modeling the luminaire's near-field environment. It would appear that a fundamentally different approach is needed.

Luminance and Helios

Modeling the near-field environment implies that we are interested in the illuminance of a point on a surface in that environment. In other words, we need to determine the luminous flux per unit area incident upon the surface at the given point, or:

$$E = dF / dA \quad (1)$$

where E is the illuminance, F is the luminous flux and dA is the differential area surrounding the point.

The illuminance of a point can also be described in terms of luminance, which is defined as:

$$L = dF^2 / (d\omega dA \cos(\theta)) \quad (2)$$

where L is the luminance, F is the luminous flux propagated in a given direction (Figure 1), $d\omega$ is the differential solid angle of an elemental cone containing the direction, dA is the differential area of the surface surrounding the point, and θ is the angle between the elemental cone and the surface normal N .

The relation between luminance and illuminance is thus:

$$E = \int_{\omega} L \cos(\theta) d\omega \quad (3)$$

We commonly think of luminance as a property of the physical surface that is emitting or reflecting light. However, the formal definition of luminance (ANSI/IES [1986]) includes both real and imaginary surfaces, where the luminous flux may be "leaving, passing through and/or arriving at the surface."

While ANSI/IES [1986] defines luminance in terms of a surface, there are physical light sources that do not have definable surfaces. One example is a plasma arc, which emits light throughout its volume rather than solely from its surface. Another example is the sky -- the blue light we see is due to the scattering of sunlight by air molecules and dust from ground level to the upper reaches of the atmosphere. These are *volume* sources of light.

ANSI/IES [1986] addresses volume sources by stating that a surface can be real or imaginary. In other words, one can choose an arbitrary imaginary plane and in theory measure its luminance.

Moon and Spencer [1944] addressed the issue more directly by introducing their "helios" concept:

"Helios is a sort of generalized brightness [luminance], which in many cases reduces to the older concept. Helios is not a characteristic of a surface but is a condition at a point in the light field. Its value depends on the *position* of the point and on the *direction* in which helios is specified.

"The helios H at a given point P and in a given direction is defined as π times the luminous flux density in that direction per unit solid angle."

Moon and Spencer proposed helios within the framework of a photometric and radiometric theory that was based solely on physically measurable properties. While correct, their arguments were largely tautological in that their measurable properties of "pharosage" and "helios" are illuminance and luminance (defined for a receiving surface) by another name (Levin [1968]).

Nevertheless, the underlying *concept* of helios remains valid: luminance is an intrinsic property of the field of light surrounding an observer in space. It is *not* a property of any surface, real or imaginary.

Levin [1968] offered a useful geometrical interpretation of luminance by noting that an elemental cone is equivalent to a geometric ray. Thus, luminance is an intrinsic property of a directed geometric ray of light. These rays are familiar from geometrical optics. As such, they are straight lines only in an optically homogeneous medium, and may be reflected or refracted at media boundaries.

The luminance of a ray is invariant in a non-participatory medium such as a vacuum or (over short distances) air. In other words, the luminance of a ray may vary as it traverses a volume source or a medium in which absorption, scattering or dispersion occurs. In a vacuum or air, however, the measured luminance remains constant.

The Helios Approach

Moon and Spencer's helios concept is directly applicable to the near-field problem. The illuminance of a point on a surface is due to the sum of the luminances of each geometric ray incident upon the surface at that point, times the cosine of the angle of each ray with the surface normal. If we know the luminance of each ray in the environment, we can calculate the illuminance at any point on any surface anywhere within the three-dimensional space surrounding the luminaire.

If we think of each ray as a bound vector, we can consider the spatial distribution of luminous flux surrounding a luminaire as being a three-dimensional vector field of light. (The analogy is mathematically inexact -- it is actually a five-dimensional scalar field, the "photic field" of Moon and Spencer [1981]. Nevertheless, the analogy is conceptually useful.) The helios approach to near-field photometry involves measuring this field rather than the surface luminance or any other intrinsic property of the luminaire.

One interesting aspect of the helios approach is that we do not need to know anything about the geometry of the luminaire or its distance from the surface being illuminated in order to calculate the illuminance at a point. (If we know the reflectance distribution function of the surface, we can also calculate its luminance.) The source (or sources) of luminous flux can even be outside the volume of space whose photic field has been measured.

Another interesting aspect is that for any point within the volume of space, we can translate our knowledge of the photic field into a photometrically accurate view of the environment as seen from that point. This has interesting implications for computer graphics. Unlike approaches to near-field photometry based on point sources (e.g., Verbeck and Greenberg [1984]), we do not require any knowledge of the objects in the environment.

The helios approach is completely general in that it can in theory model (within the limits of geometrical optics) the spatial distribution of luminous flux surrounding a luminaire with absolute accuracy and precision. In doing so, it fully characterizes both the near-field and far-field environment. The essential problem, of course, is that of measuring the photic field.

Photic Field Measurement - Theory

Measuring the photic field surrounding a luminaire in general requires measuring the luminance of every possible ray at every point in three-dimensional space. This is necessary, however, only for participatory media. For media where absorption, scattering and diffusion are negligible (e.g., air), we can utilize the principle of luminance invariance.

Figure 2 shows a luminaire L illuminating a surface S and several directed geometric rays. Those that originate within the luminaire may have non-zero luminance; all other rays have zero luminance.

Consider ray #2, which originates with the luminaire and intersects the surface. The ray luminance within the luminaire (which is in general a volume source) will vary along its length. We do not know the path of the ray within the luminaire or where it begins. What we do know is that between the luminaire and the surface, the ray is a straight line and its luminance is constant.

If we remove the surface, rays #2 and #3 extend from somewhere within the luminaire to infinity. Their luminance is constant from the luminaire (which may have an ill-defined surface) to infinity. We may measure this luminance at any convenient point along the rays. In general, we can do this for any ray, whether or not it originates within the luminaire or intersects the surface.

Figure 3 shows the same luminaire inside a convex closed surface C . We can in theory measure the luminance of every ray intersecting this surface. In doing so, we measure and fully characterize the photic field surrounding the luminaire (Levin [1971]), both inside *and outside* the closed surface.

It is important to recognize that the closed surface is *imaginary*. It simply represents the (infinite) set of positions at which the luminance of each ray (of which an infinite number intersects each position) is measured.

What is interesting about this technique is that it allows a set of photometric measurements to be made at a fixed distance from the luminaire. The resultant data, however, can be used to calculate photometric quantities at any distance and in any direction (Figure 4).

There is an obvious parallel between this technique and that of far-field photometry. If the luminaire is a point source, then there is only one possible ray that originates within the luminaire and intersects a given position on the imaginary closed surface. The far-field photometer measures illuminance, which in this case is the luminance of the ray. Far-field photometry is thus the limit of the helios approach to near-field photometry.

There are no restrictions on the geometry or physical properties of the surface for which point-by-point illuminance predictions are to be made. The surface can be oriented at any angle with respect to the luminaire at any given point. It does not have to be planar or continuous. The point can be located inside or outside the imaginary closed surface defined by the ray luminance measurement positions.

There are two minor exceptions. One is that the point cannot be located inside the convex hull of the luminaire. That is, any geometric ray originating within the luminaire

volume must not intersect the volume after passing through the point (Figure 5). If the surface of the luminaire is ill-defined, an imaginary *bounding volume* can be defined which fully encloses the luminaire; the point cannot be located within this volume.

The second exception is common to all photometric measurement techniques: it must be assumed that the surface being illuminated does not reflect luminous flux back into the luminaire volume. If it does, then the luminaire becomes photometrically coupled with its environment, and photometric measurements must be performed in situ.

Photic Field Measurement - Practice

To be useful, the helios approach requires an instrument capable of quickly and accurately measuring the luminance of geometric rays. We also must establish what finite subset of the infinite number of rays is sufficient to characterize the photic field surrounding a typical luminaire.

The first consideration is the minimum distance of the luminance meter from the luminaire. Far-field photometry is generally performed such that the illuminance meter is no closer than five times the maximum width of the luminaire (IESNA [1985]). In near-field photometry, the luminance meter can be placed anywhere outside the bounding volume of the luminaire. In practice, a distance of approximately the maximum width of the luminaire is convenient. For a typical 1219 mm (4-foot) fluorescent luminaire, this translates to a distance of 1.2 meters from the luminaire center.

The next question concerns the luminance meter design. It is clear that we cannot directly measure the luminance of an infinitesimal geometric ray; we are limited to measuring the luminous flux contained within a finite solid angle and incident upon the finite area of a photosensor. As shown in Figure 6a, this describes an aperture-type luminance meter. The volume of space it measures is only an approximation of a geometric ray. If we accept that far-field photometry adequately models the far-field environment (that is, at distances greater than five times the maximum luminaire width), we can choose an approximation that is only valid within the near-field environment.

A typical commercial spot luminance meter has an optical system (Figure 6b) to focus an image of the surface of interest onto its photosensor. Its field of view when focused at infinity is about one degree, and its optical system has a clear aperture of 40 mm or so. As an approximation to a geometric ray, this represents a truncated cone with a minimum diameter of 40 mm and a solid angle of one degree. The lens can be stopped down to provide a smaller clear aperture, subject only to photosensor sensitivity and diffraction effects. In short, the meter's field of view offers a good approximation of a geometric ray.

A lens-type luminance meter can be focused at any distance, including that which places the measurement plane inside the luminaire volume. To yield meaningful results, it should be focused on a plane between the luminaire and the meter. When focused at infinity, the measurement plane is the surface of the meter photosensor. This focus position also results in the smallest solid angle for the field of view.

There remains the problem of orienting the luminance meter at each position to measure the luminance of each ray intersecting that point in space. While a motor-driven gimbal mount or optical scanners could be used, there is a much more practical alternative: the CCD multielement photosensor.

A Compound Luminance Meter

Rea and Jeffrey [1990] and Lewin et al. [1991] have demonstrated that a suitably calibrated CCD-based video camera and accompanying frame grabber is capable of performing highly accurate point-by-point *surface* luminance measurements. By focusing the camera at infinity, it is equally capable of performing accurate ray luminance measurements. Each CCD photosensor element (called a *pel*) measures the luminance of a single geometric ray (actually a pyramidal cone).

If the CCD camera's field of view is 90 degrees and the CCD sensor is an array of 512 x 512 pels, the field of view of each CCD pel has a solid angle of only 0.2 degrees, making it an excellent approximation of a geometric ray. The precise geometry of the photosensor (accurate to within fractions of a micron) ensures that the orientation of any ray within the camera's field of view can be determined with arc-minute accuracy. Since each captured video frame simultaneously measures the luminance of over 250,000 rays, an appropriate term for the instrument is a *compound luminance meter*.

It is important to note that by focusing the camera at infinity, the luminaire is likely placed out of focus. As shown in Figure 7, this is a consequence of Moon and Spencer's interpretation of luminance. Each CCD pel is "imaging" a volume of space rather than a specific surface. Ignoring losses within the optical system, the luminous flux measured by each pel is exactly the average luminance of that volume. (It is still convenient, however, to refer to the camera's "view" of the luminaire.)

A practical apparatus for performing ray luminance measurements is shown in Figure 8. The compound luminance meter is fastened to a rigid arm that is rotated in the vertical plane about the luminaire. If the meter's overall field of view encompasses the bounding volume of the luminaire, the meter can be rigidly fastened to the arm and pointed at its center of rotation. The luminaire itself is rotated about its vertical axis, thus allowing the meter to be positioned anywhere on the surface of an imaginary sphere surrounding the source.

Rea and Jeffrey [1990] detailed numerous techniques to ensure the photometric accuracy of their CCD-based luminance meter, including modifying the spectral response with a V-lambda correction filter, disabling the automatic gain control (AGC) and gamma correction circuitry in the camera, subtracting dark frame measurements and averaging the readings of several captured video images. Lewin et al. [1991] found that the camera performance can be affected by temperature and humidity fluctuations.

CCD photosensors are inherently linear devices. However, their dynamic range is typically in the range of 45 dB, due primarily to thermally induced noise. While this performance can be improved by cooling the CCD sensor, we have instead chosen to operate the sensor within its normal dynamic range. As shown in Figure 8, we have placed a beam splitter in front of the camera that provides the same field of view to both the camera and a calibrated illuminance meter. The illuminance meter provides an absolute photometric reading for each camera position, against which the CCD sensor readings (ranging from 0 to 255 units) are scaled.

A secondary advantage of scaling the CCD sensor readings to the illuminance meter readings is that the camera's AGC does not have to be disabled. Rea and Jeffrey [1990] adjusted the lens aperture of their camera to ensure that the light source image was within the CCD sensor's dynamic range. However, they found that it was necessary to compensate for vignetting effects on a per-pixel basis. This compensation had to be predetermined for each combination of lens aperture and focal length. If the camera's AGC is enabled, the lens aperture can remain fixed as the camera rotates about the luminaire.

Data Storage Considerations

An initial concern during the development of the apparatus was that storage of the video images would consume many megabytes of data, even for a single luminaire. (Each 512 x 512 video frame represents 256 kilobytes of data.) Fortunately, this has not been the case. We have determined (see *Experimental Results* below) that image resolutions of 64 x 64 and even 32 x 32 are sufficient for near-field illuminance calculations. These resolutions represent 4,096 and 1,024 bytes respectively per video frame.

A useful side effect occurs in that each pixel in a 64 x 64 image is the average of an 8 x 8 block of pixels from the original video frame. The signal-to-noise (S/N) ratio of the image is improved by the square root of the number of pixels per block -- a factor of eight. Whereas Rea and Jeffrey [1990] had to sum 32 video frames to achieve an acceptable S/N ratio for their 512 x 480 images, a single 64 x 64 image derived from a 512 x 512 video frame is equivalent to summing 64 frames; a 32 x 32 image is equivalent to summing 256 frames.

We also found that angular increments of 5 degrees between meter positions in both the vertical and horizontal planes are sufficient for most near-field calculations. For a luminaire with quadrilateral symmetry, this implies $19 \times 37 = 703$ video frames. However, we need fewer images in the horizontal planes for vertical angles near nadir and zenith. In practice, only 540 images need to be stored. This represents approximately 2.1 megabytes of data for a typical luminaire at an image resolution of 64×64 .

Fortunately, any video image has a considerable amount of intraframe (spatial) redundancy. Also, since each video frame presents a slightly different view of the luminaire from that of its neighbors, there is a considerable amount of interframe (temporal) redundancy as well. This redundancy can be removed using a variety of image compression techniques.

There are two classes of image compression: lossless and lossy. Lossless compression means that the original data can be recovered from the compressed data with no loss of accuracy. Raw video image data files can typically be compressed by a factor of 4 to 20. This implies a near-field photometric file size of 100 to 500 kilobytes.

Lossy compression means that some of the accuracy of the original data is sacrificed when the data is compressed. The resultant errors are typically distributed over the entire video frame. Typical compression factors for error ratios of 30 dB or better can range from 20 to 2500, depending upon the image complexity. Since we are scaling the video frame data to independently determined illuminance values, lossy image compression techniques appear to be applicable to near-field photometry. Data file sizes of 5 to 100 kilobytes are likely possible.

We are currently implementing several lossless and lossy image compression techniques, including the ISO/CCITT JPEG still picture and forthcoming MPEG moving picture image compression standards (Wallace [1991] and Le Gall [1991]) and techniques based on iterated contractive functions (Jacquin [1990]) and wavelet transforms (Pentland and Horowitz [1991]). Using these techniques, we fully expect to be able to provide between 3 and 100 near-field photometric data files on a single 3-1/2 inch floppy disk.

Illuminance Calculations

Having described a practical near-field goniophotometer, it remains to describe an efficient illuminance calculation method based on the helios approach. The calculation method presented here derives many of its features from radiative transfer theory (e.g., Sparrow and Cess [1978]) and the radiosity method of Cohen and Greenberg [1985] for modeling complex environments.

Figure 9 shows an elemental convergent cone $d\omega$ (a geometric ray) incident upon a point P on a surface at an angle θ . The illuminance of the point is due to all such rays incident upon the surface, and is given by (3) above.

Each cone $d\omega$ intersects the imaginary unit hemisphere enclosing the point. The orthogonal projection of the differential area of intersection onto the surface, dC is given by:

$$dC = \cos(\theta) d\omega \quad (4)$$

From Nusselt's analogy, dC is the differential configuration factor of $d\omega$, and represents the fraction of luminous flux in $d\omega$ that is intercepted by the differential surface area surrounding the point (Cohen and Greenberg [1985]). The illuminance of the point is then given by:

$$E = \int_C L dC \quad (5)$$

We can approximate the illuminance of the point by grouping the elemental convergent $d\omega$ cones into a set of finite convergent cones ω . The luminance L_i of each finite cone ω_i is the average of its constituent elemental cones. The approximate illuminance E' is then given by:

$$E' = \sum_i L_i \Delta C_i \quad (6)$$

where ΔC_i is the delta configuration factor of cone ω_i .

Cohen and Greenberg [1985] substituted a *hemi-cube* for Nusselt's hemisphere (Figure 10). It has computational advantages in that each face can be divided into square elements ΔA , with delta configuration factors given by:

$$\Delta C_{\text{top}} = \Delta A / (\pi (x^2 + y^2 + 1)^2) \quad (7)$$

and

$$\Delta C_{\text{side}} = z \Delta A / (\pi (y^2 + z^2 + 1)^2) \quad (8)$$

(The derivation of these formulae is presented in Cohen and Greenberg [1985].)

The illuminance calculation method thus proceeds as follows: given a point on a surface in the near-field environment surrounding a luminaire, a hemi-cube is centered over the

point. A geometric ray is projected from the point out through the center of each hemi-cube element into the environment. Only those rays that intersect the bounding volume of the luminaire need be considered.

If the point is located outside the imaginary sphere defined by the luminance meter positions (the *measurement sphere*), the intersection of the ray with the sphere (i.e., its position and orientation) is calculated. Otherwise, the ray is extended through the point in the opposite direction until it intersects the measurement sphere (Figure 4).

The hemi-cube ray will likely not intersect the measurement sphere exactly at a meter position or with the exact orientation of a measured ray. The closest rays that were measured must be determined, and the hemi-cube ray luminance calculated using bilinear interpolation. Having determined each ray's luminance, the illuminance of the point can be determined from (6) above.

The illuminance calculation method may be expressed in pseudocode as:

Point illuminance $E' = 0$

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FOR each hemi-cube element ray  $r_i$ 
  IF  $r_i$  intersects luminaire bounding volume
    Determine intersection with measurement sphere
    Calculate luminance  $L_i$  using bilinear interpolation
    Calculate element configuration factor  $\Delta C_i$ 
     $E' = E' + L_i * \Delta C_i$ 
  ENDIF
ENDFOR

```

An efficient implementation of this method must consider a number of related issues, including near-field distance limits, 3-D co-ordinate transformations, hidden surface calculations, supersampling of hemi-cube elements and video frame pixels, photometric data file structures, video frame cache algorithms and file compression techniques. Most of these issues have been thoroughly documented in the computer graphics and image processing literature.

A Simplified Approach

The above illuminance calculation method can predict the illuminance at any point on any surface anywhere within the 3-D space surrounding the luminaire. Unfortunately, it must trace a large number of rays in order to do so. While the calculations for a single point can be performed in a reasonable amount of time on a desktop computer (see *Experimental Results* below), the method becomes unwieldy when point-by-point

illuminance calculations for a plane are required -- which is the commonest situation for interior lighting calculations.

Fortunately, the major surfaces of most architectural interiors are orthogonal to one another and aligned with the luminaires. That is, the ceiling and floor planes are usually parallel to the two horizontal axes of the luminaires, and the wall and room partition planes are usually parallel to the luminaires' vertical axis and one of their horizontal axes.

Taking a suspended indirect fluorescent luminaire as an example, we can imagine a set of parallel ceiling planes arranged at various distances above the luminaire (Figure 11). The point-by-point illuminance of each plane can be calculated using the above calculation method. If we store these calculations in a data file, the point-by-point illuminance of a ceiling plane at any distance can be determined either directly or by interpolation between two precalculated planes.

Similar calculations could be performed for the floor and wall planes. A typical luminaire with quadrilateral symmetry would require two sets of wall plane calculations; bilateral symmetry would necessitate three sets.

This approach bears some resemblance to that proposed by Ngai et al. [1991] for application distance photometry. It has an advantage, however, in that the point-by-point illuminance of a large number of parallel planes can be calculated from a single photometric data set. A subset of these planes can then be chosen for storage such that point-by-point illuminances for intermediate planes can be interpolated with a guaranteed error limit.

A complete set of point-by-point illuminances for three or more sets of parallel planes for a typical luminaire will likely consume several megabytes of file space. Fortunately, there are several opportunities for lossless or lossy data compression. The data to be stored is similar in form to that used in *volume visualization* applications such as computer-aided tomography (e.g., Kaufman [1991]). We are currently investigating various data compression techniques that are in use or have been proposed for such applications, including octree encoding, chain codes and 3-D fractal image compression.

Visualization

Cohen and Greenberg [1985] introduced the hemi-cube as an efficient method for computing form factors. One can imagine, however, looking out from the point through any of its faces. Each element has a luminance equal to that of the finite cone it intercepts. A graphical display of the hemi-cube faces (e.g., Figures 12A-C) is therefore a synthesized view of the environment as seen from the point. The displayed image is photometrically accurate to within the rendering capabilities of the display device.

One useful property of the hemi-cube is that it offers a full 180-degree field of view of the environment without any nonlinear distortion through each face. Straight lines in the environment project as straight lines onto the hemi-cube faces, making displayed images of luminaires easy to interpret. A fish-eye lens can also display a 180-degree field of view, but only with severe distortion. If needed, the five hemi-cube faces can be mapped into a fish-eye view using a simple co-ordinate transformation (Hill [1990]).

Experimental Results

The photometric accuracy of the near-field goniophotometer is dependent upon the accuracy of the CCD photosensor array and associated electronic and optical components. This apparatus is currently under development in the author's laboratory. Its components are similar to those employed by Rea and Jeffrey [1990].

Preliminary experiments were performed in co-operation with the National Research Council of Canada using the CapCalc luminance meter developed by Rea and Jeffrey. The primary purpose of these experiments was to evaluate the suitability of CCD video cameras for near-field photometry. Each 512 x 480 video image consisted of 960 kilobytes of data (one 4-byte floating point number for each pixel). It was demonstrated that an average lossless image compression ratio of 16:1 could be achieved using a Ziv-Lempel encoder algorithm.

The CapCalc system was used to perform near-field and far-field measurements of a quartz-halogen MR-16 narrow beam lamp. The camera was positioned at a distance of 2.0 meters from the lamp center and measurements taken at 2.5 degree intervals in a single vertical plane about the lamp axis. A 22 kilobyte data file comprising 73 run-length encoded 64 x 64 resolution images was prepared and used to predict the near-field illuminance distribution of planes spaced from 55 to 80 mm from the lamp center. Unfortunately, a number of experimental uncertainties prevent any quantitative conclusions to be drawn from the measured versus predicted illuminance values. We are currently awaiting completion of the near-field goniophotometer to perform useful tests on physical luminaires.

Theoretical Results

The prediction accuracy of the illuminance calculation method is dependent on the number of discrete views available and the luminance meter image resolution. Since these parameters are independent of the photometric accuracy of the CCD measurements, we created two synthetic photometric data sets for a circular luminous disk with a uniform Lambertian luminance distribution of 100 cd/m². One set had an equivalent luminance meter resolution of 64 x 64 pixels, the other had a resolution of 32 x 32 pixels. Thirty-seven views of the disk were synthesized over 5 degree increments in the vertical

plane. Since a disk is rotationally symmetric, data for only one vertical plane was required.

The advantage of such a data set is that the predicted illuminances for any distance and orientation can be compared with those predicted by radiative transfer theory. Two cases were considered: (1) an illuminated surface parallel to the disk, and (2) an illuminated surface perpendicular to the disk.

The disk diameter was set at 1000 mm. For the parallel case, the illuminated surface was set at distances of 125, 250, 500, 1000 and 2000 mm from the disk plane. The calculation point was located at distances ranging between 0 and 3333 mm from the disk origin.

For the perpendicular case, the illuminated surface was set at distances of 500, 555, 714 and 1000 mm from the disk origin. The calculation point was located at distances ranging between 50 and 3333 mm from the disk plane.

A theoretical illuminance prediction for each point was calculated in accordance with formulae presented in Sparrow and Cess [1978]. Two numerical predictions based on the method outlined above (see *Illuminance Calculations*) were calculated, one for each data set. The hemi-cube was divided such that there were 128 x 128 elements on the top face and 64 x 128 elements on each side. The calculation program recursively subdivided each hemi-cube element into (and traced rays through) as many as 64 subelements, depending on an estimate of the bounding volume's configuration factor.

A subset of the results is shown in Tables 1 through 4. The relative and total errors (defined in the tables) are typical for all cases examined. The large relative errors for the furthest point distances (parallel case) are due to the small configuration factors involved -- only a limited number of hemi-cube rays actually intersected the disk. The accuracy could have been improved by increasing the number of hemi-cube subelements.

For the majority of the point and plane distances, the numerical and theoretical predictions agree to within ± 1 percent for the 64 x 64 data set, and to within ± 2 percent for the 32 x 32 data set. This implies that a luminance meter resolution of 32 x 32 is likely adequate for most near-field applications.

Our illuminance calculation program included the ability to graphically display the hemi-cube faces, where the relative luminance of each hemi-cube element was indicated by one of 16 colors. Synthetic 64 x 64 data sets for luminous disks consisting of concentric rings with various luminances were created to test the ability of the calculation program to synthesize views of the luminaire. A maximum of 16 equally spaced rings could be visually resolved. This offers empirical evidence that the illuminance calculation method is capable of synthesizing views of the near-field environment to within the resolution limits imposed by the Sampling Theorem.

The synthesized views of the luminous disk are not perfect. For example, the images are geometrically distorted when the calculation point is very close to the disk. This distortion is due primarily to the limited number of views in the data set.

Another problem appears when the disk is viewed directly from above at a distance equal to the disk radius. The geometry of the disk is preserved, but the luminance distribution at the edge of the disk is radially distorted. The effect is minor in terms of the illuminance prediction errors, as evidenced by Tables 1 and 2. The problem is due to aliasing, and can be alleviated by increasing the number of discrete views.

We developed our illuminance calculation program as a prototype to validate the calculation method algorithm. At present, a single illuminance calculation requires between 2 and 25 seconds on a 25 MHz 386-class desktop computer. We expect to improve this performance by a factor of at least four using a variety of software optimization strategies.

Conclusions

Current lighting calculation practices based on far-field photometry do not accurately model the near-field environment of a luminaire. The need for accuracy has been clearly stated by Mistrick and English [1990] and others. The influence of computer visual display terminals on office lighting design (IESNA [1989]) has further emphasized this need.

This paper has presented a new approach to near-field photometry and described a novel near-field goniophotometer and illuminance calculation method. The approach has been shown to be completely general in that it fully characterizes the near-field environment. The goniophotometer uses existing CCD video technology and is inexpensive to manufacture. The calculation method is computationally efficient and accurate, and can be implemented in software for today's desktop computers.

The helios approach to near-field photometry has implications beyond that of improved lighting calculations. The illuminance calculation method can be merged with ray tracing and radiosity techniques to further improve the realism of computer-generated images. Luminaires can be displayed in detail, and shadows can be accurately portrayed at any distance. The approach allows us to both calculate and visualize the near-field environment of any luminaire.

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Tables

Light source:	Synthetic luminous disk 1000 mm diameter 100.0 cd/m ² Lambertian luminance
Data File:	One vertical plane 37 CCD images at 5-degree increments
Notes:	Pred E = predicted illuminance Theor E = theoretical illuminance Relative error = (Pred E - Theor E) / Theor E Total error = Pred E - Theor E x = parallel distance from disk origin y = perpendicular distance from disk origin $D = y / x$

Table 1

Orientation: Plane parallel to disk
 CCD image resolution: 64 x 64 pixels
 Perpendicular distance: 250 mm

Parallel	D	Theor E	Pred E	Rel Error	Total Error
3333 mm	0.15	0.0131	0.0112	-14.7 %	0.0 %
2500	0.20	0.0425	0.0395	-6.9	0.0
2000	0.25	0.1072	0.1028	-4.1	0.0
1667	0.30	0.2311	0.2167	-6.2	0.0
1429	0.35	0.4482	0.4358	-2.8	0.0
1250	0.40	0.8065	0.7829	-2.9	0.0
1000	0.50	2.2110	2.1675	-2.0	0.0
833	0.60	5.1661	5.1230	-0.8	0.0
714	0.70	10.4923	10.3854	-1.0	-0.1
625	0.80	18.4704	18.3185	-0.8	-0.2
555	0.90	28.2866	28.0688	-0.8	-0.2
500	1.00	37.8732	37.8049	-0.2	-0.1
333	1.50	64.6498	64.4261	-0.3	-0.2
250	2.00	72.3607	71.8760	-0.7	-0.5
200	2.50	75.4074	75.2473	-0.2	-0.2
167	3.00	76.9061	76.9057	0.0	0.0
143	3.50	77.7792	77.8888	0.1	0.1
125	4.00	78.3264	78.7393	0.5	0.4
100	5.00	78.9461	79.4836	0.7	0.5
83	6.00	79.2804	79.6687	0.5	0.4
71	7.00	79.4763	79.7213	0.3	0.2
63	8.00	79.5889	79.7314	0.2	0.1
56	9.00	79.6760	79.7674	0.1	0.1
50	10.00	79.7422	79.7735	0.0	0.0
0	INF	80.0000	79.8933	-0.1	-0.1

Table 2

Orientation: Plane parallel to disk
 CCD image resolution: 32 x 32 pixels
 Perpendicular distance: 250 mm

Parallel	D	Theor E	Pred E	Rel Error	Total Error
3333 mm	0.15	0.0131	0.0172	-31.6 %	0.0 %
2500	0.20	0.0425	0.0535	-26.0	0.0
2000	0.25	0.1072	0.1134	-5.8	0.0
1667	0.30	0.2311	0.1935	-16.2	0.0
1429	0.35	0.4482	0.4047	-9.7	0.0
1250	0.40	0.8065	0.7563	-6.2	-0.1
1000	0.50	2.2110	2.0804	-5.9	-0.1
833	0.60	5.1661	4.9608	-4.0	-0.2
714	0.70	10.4923	10.1196	-3.6	-0.4
625	0.80	18.4704	17.7875	-3.7	-0.7
555	0.90	28.2866	27.2574	-3.6	-1.0
500	1.00	37.8732	36.9662	-2.4	-0.9
333	1.50	64.6498	63.6250	-1.6	-1.0
250	2.00	72.3607	70.8804	-2.0	-1.5
200	2.50	75.4074	74.3758	-1.4	-1.0
167	3.00	76.9061	76.1024	-1.0	-0.8
143	3.50	77.7792	77.4423	-0.4	-0.3
125	4.00	78.3264	78.1990	-0.2	-0.1
100	5.00	78.9461	78.8146	-0.2	-0.1
83	6.00	79.2804	78.9443	-0.4	-0.3
71	7.00	79.4763	78.9132	-0.7	-0.6
63	8.00	79.5889	79.0165	-0.7	-0.6
56	9.00	79.6760	78.9658	-0.9	-0.7
50	10.00	79.7422	78.7426	-1.3	-1.0
0	INF	80.0000	78.5381	-1.8	-1.5

Table 3

Orientation: Plane perpendicular to disk
 CCD image resolution: 64 x 64 pixels
 Parallel distance: 555 mm

Perpend	D	Theor E	Pred E	Rel Error	Total Error
3333 mm	0.15	0.3403	0.3391	-0.4 %	0.0 %
2500	0.20	0.7521	0.7519	0.0	0.0
2000	0.25	1.3507	1.3477	-0.2	0.0
1667	0.30	2.1213	2.1319	0.5	0.0
1429	0.35	3.0371	3.0469	0.3	0.0
1250	0.40	4.0673	4.0576	-0.2	0.0
1000	0.50	6.3246	6.3144	-0.2	0.0
833	0.60	8.6767	8.5987	-0.9	-0.1
714	0.70	10.9688	10.9682	0.0	0.0
625	0.80	13.1217	13.0454	-0.6	-0.1
555	0.90	15.1318	15.0262	-0.7	-0.1
500	1.00	16.9330	16.9380	0.0	0.0
333	1.50	23.7437	23.6104	-0.6	-0.1
250	2.00	27.8341	27.6363	-0.7	-0.2
200	2.50	30.3598	29.9892	-1.2	-0.4
167	3.00	31.8928	31.4274	-1.5	-0.5
143	3.50	32.8046	32.1699	-1.9	-0.6
125	4.00	33.2649	32.6235	-1.9	-0.6
100	5.00	33.3027	32.4956	-2.4	-0.8
83	6.00	32.6111	31.4649	-3.5	-1.1
71	7.00	31.5291	30.1974	-4.2	-1.3
63	8.00	30.4122	29.1040	-4.3	-1.3
56	9.00	29.0950	27.8021	-4.4	-1.3
50	10.00	27.6570	26.3720	-4.6	-1.3

Table 4

Orientation: Plane perpendicular to disk
 CCD image resolution: 32 x 32 pixels
 Parallel distance: 555 mm

Perpend	D	Theor E	Pred E	Rel Error	Total Error
3333 mm	0.15	0.3403	0.3385	-0.5 %	0.0 %
2500	0.20	0.7521	0.7438	-1.1	0.0
2000	0.25	1.3507	1.3412	-0.7	0.0
1667	0.30	2.1213	2.1184	-0.1	0.0
1429	0.35	3.0371	2.9989	-1.3	0.0
1250	0.40	4.0673	3.9889	-1.9	-0.1
1000	0.50	6.3246	6.3346	0.2	0.0
833	0.60	8.6767	8.6130	-0.7	-0.1
714	0.70	10.9688	10.8300	-1.3	-0.1
625	0.80	13.1217	13.0666	-0.4	-0.1
555	0.90	15.1318	15.0485	-0.6	-0.1
500	1.00	16.9330	16.9581	0.1	0.0
333	1.50	23.7437	23.2201	-2.2	-0.5
250	2.00	27.8341	27.5986	-4.4	-1.2
200	2.50	30.3598	29.1664	-3.9	-1.2
167	3.00	31.8928	30.3657	-4.8	-1.5
143	3.50	32.8046	30.9380	-5.7	-1.9
125	4.00	33.2649	31.0923	-6.5	-2.2
100	5.00	33.3027	31.2349	-6.2	-2.1
83	6.00	32.6111	30.5475	-6.3	-2.1
71	7.00	31.5291	29.4499	-6.6	-2.1
63	8.00	30.4122	28.1262	-7.5	-2.3
56	9.00	29.0950	26.4087	-9.2	-2.7
50	10.00	27.6570	24.4278	-11.7	-3.2

Discussion

The author is to be congratulated on having spent the time and effort expended in the investigation of the procedure and the preparation of this paper. It is a good start.

There is a great deal of work to be done in tracking down and correcting the errors that were evident in the representation of the Lambertian disk, and real life luminance distributions have not yet been considered.

All of these procedures, whether they be intensity measurements and computer prediction of resulting luminances and illuminances are nothing more than attempt to quantify the subjective evaluation of the human eyeball. The more sophisticated the attempt to quantify or represent, the greater chances there are for errors to creep in and render the whole thing meaningless.

I am not belittling the effort or the knowledge gained by the preparation of this work, rather I am evangelizing a return to the KISS (an acronym familiar to all) philosophy. The simpler something is, the easier it is to apply, understand, explain and use.

Thomas L Ballman, FIES
Building Acoustics and Lighting Laboratories

The concept proposed by the author for near-field photometry is a very good one. When it is implemented for its original intent, it has some important advantages over the concept proposed by the discussor. Specifically, the ease of handling intervening surfaces between the light source and the illuminated surface. The other advantage is its ability to handle non-diffusing surfaces. The discussor has investigated this technique first with photographic photometry, and then with CAPCALC when it was introduced a few years ago.

The author's conclusion is that it is extremely difficult to obtain luminances of an actual "scene" with the degree of accuracy that is demanded by photometric testing. The difficulties are many. Some have to do with non-theoretically perfect optical lenses; some have to do with CCD photosensors themselves. Some of these difficulties can be overcome by extraneous and careful calibration; others cannot.

One of the most elusive problems seems to be the "point spread" nature of the optical train. Hence, the intensity received by an individual pel is not only dependent on the specific "cone of light", but is also influenced by the adjacent intensities. It cannot be calibrated out because adjacent intensities vary.

In terms of computational complexity, the discussor has done extensive work in the technique the author mentioned. It is by using this method that one can overcome the problem of intervening and non-diffusing surfaces. The difficulty in using this technique

is the time it requires. Normally, realistic point-by-point calculations for a realistic lighting layout is on the order of minutes. Using the technique proposed by the author it will take anywhere from six hours to more than 24 hours using a present-day desktop computer.

The simplified approach proposed by the author is a good one. However, by using this simplified approach the author has forfeited the two major advantages, namely, again, the solutions to the problems of intervening surfaces and non-specularity calculation.

In summary, this is an approach the discussor considered, but was unable to apply in practice in a day-to-day photometric testing environment. It is the discussor's sincere hope that the author can carry on this work and derive a practical solution to the obstacles encountered in this approach.

Peter Ngai
Peerless Lighting

The paper suggests that the attribute of quadrilateral symmetry will require fewer video images to characterize a luminaire's distribution. For such luminaires, current far-field practice does not provide for a reduced volume of raw data. Rather, measurements are made over the entire luminaire -- the final reduced candela set then results from the averaging of the measurements into a single quadrant. CCD measurements should likewise be made of the entire luminaire, so that the number of video images would not be less than with a completely asymmetric luminaire. However, the reduced final data set could represent a single quadrant and thus be smaller than its asymmetric counterpart.

It is worth emphasizing that the author's method for calculation of the illuminance is in a sense the "inverse" of the methods used in most of the computer software available today for that purpose. This is because the paper's approach is to fix attention on the target point and in effect ask "Where is the light coming from, and what is its magnitude?" By contrast the traditional approach is to fix attention on the luminaire and ask "How much light is going to the target point?"

Here is a suggestion for what seems to this discussor a more elegant and aesthetically appealing technique than the use of the hemi-cube: center a hemisphere (rather than a hemi-cube) over the calculation point. Partition the surface of the hemisphere into 'n' regions such that each region has the same configuration factor. For each region, trace the ray which passes through its center from the point. As we move from the hemisphere's equator towards its zenith, the regions become smaller. In a sense the "resolution" is therefore coarser for rays near the equator. For most cases of interest this is beneficial because rays leaving the hemisphere near the equator will not intersect the luminaire anyway. The expensive calculation is done in advance and its results stored for lookup -- i.e., the actual partitioning of the sphere and the resulting (x,y,z) location of

each exiting ray. At computation time the actual ray to be traced is determined by a 3-d transformation of the prestored ray. Each configuration factor is simply $1 / 'n'$.

In addition to those pointed out by the author, the use of this technique involves an additional uncertainty relative to conventional far-field photometry. For illuminance calculations with conventional photometry one must know the photometric distribution and the relative locations of the luminaire and the calculation point. With the technique presented here, one must also know the test distance used in the actual measurements. Lacking this one cannot assign measured luminances to rays being traced to determine illuminance. Any uncertainty in this parameter of course adds to the uncertainty of any computed illuminance.

In addition to its virtues in near-field applications, this method offers great benefit for far-field photometry. A dominant factor in the cost of laboratory photometric facilities is the cost of the typically large space required to house a conventional goniophotometer. If the author's method can be implemented with sufficient precision, the cost of lab facilities will drop dramatically. Other advantages of convenience will accrue also. For example, since the luminaire can be mounted at a lower height, it will be more conveniently aligned, etc. Using the paper's illumination calculation technique, the pesky "stray" / "extraneous" light problem disappears as well, for a traced ray either intersects the luminaire volume or it does not.

Compared to previous research, the author's scheme does have the advantage that the measurement time is essentially the same as for a single far-field photometric exploration (assuming that the CCD need not linger at each measurement position). This is because the earlier-suggested techniques require photometric explorations at multiple test distances. The following uncertainties must be discharged before the method can gain support as a practical laboratory procedure:

1. A satisfactory and consistent means of neutralizing the overlap inherent in the luminance measurements.
2. The sheer volume of the raw data.
3. The computer time required to reduce and/or compress the raw data.
4. The error in the finished result -- will it be within traditionally accepted limits, or is the advantage of the technique so great that the traditional limits may be relaxed?

It is not clear to this discussor that any of these are dischargeable at this time.

If the technique does survive practical implementation, it is probably the best solution yet presented for the near-field photometry problem. If as well the technique proves

satisfactory for traditional far-field photometry, then its effect will be nothing short of revolutionary.

William E. Brackett
Independent Testing Laboratories

The author proposes a photometric method which measures the spatial luminance distribution of a luminaire, in the form of a four dimensional scalar function. This notion appeared in the English literature in 1939 with the translation of Gershun's work, "The Light Field" (1). Gershun describes a "luminance distribution solid" and shows how it can be used to completely describe this scalar field. Although not commonly available in the West, Yamauti's work "Theory of Field of Illumination" (2), published in English in Japan in 1932, presaged Gershun. Although lacking the trappings of the apparatus of vectors, Levin's use of this idea is very explicit in his 1971 work (3).

What is new in this paper is the presentation of a clear way to overcome the difficulties referred to by Levin and use the scalar luminance field of a luminaire.

The use of CCD based compound luminance meters in this context is an excellent application of these devices. The use of a beam splitter and an additional monitoring illuminance meter holds the promise of providing a stable photometric procedure. This is especially helpful since the automatic gain control of a compound luminance meter can now be used.

It is not encouraging to learn of the results of the preliminary experiments. The author mentions "a number of experimental uncertainties" preventing any quantitative conclusions from being drawn. Would the author describe these experimental uncertainties?

The theoretical results reported need to be evaluated with great caution. The practicality of the system (not its correctness, which is self-evident) cannot be demonstrated with a perfectly diffusely emitting disk. Currently available luminaire modeling software (4,5) can be used to investigate the proposed system's practicality. Assuming a 90 degree field of view, the closest practical distance for the meter to a 4 foot long fluorescent luminaire is 3 feet, say. It can be shown that for many luminaires, especially those with specular reflectors, the surfaces involved require a luminance resolution of approximately 0.5 inch. This surface resolution is required to predict the large illuminance gradients produced on nearby surfaces. This corresponds to an angular resolution of approximately 0.5 degree. This, in turn, requires a pixel count of 180 x 180. Obviously, a higher pixel count will be required if the compound luminance meter is further away.

A similar analysis can be performed to investigate the angular increments between luminance meter positions. At test distances approximately equal to the size of typical

luminaires, it is not at all obvious that the 5 degree angular increments reported by the author for the diffuse disk are sufficient for real luminaires. A consideration of the geometry involved reveals an approximate inverse relationship between the angular resolution of acquired data (the number of pixels used) at each meter position, and the angular resolution (number) of meter positions.

It is not clear from the paper why a range of 256 luminance steps will be sufficient. A luminance resolution (in any one image or capture) of 1 part in 256 is insufficient, even for fluorescent luminaires. It would appear that the low luminances in such an image would need to be determined to at least 1 part in 1000, especially if the low luminances are produced by a large solid angle of the luminaire. These rough estimates of the resolutions required can be verified by the same luminaire performance modeling software mentioned above.

Presumably such details will be resolved when the author's near-field goniophotometer is operational.

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David L. DiLaura, FIES
Lighting Technologies, Inc.

This paper describes a new and innovative method to realize a conceptual process described 20 years ago to completely characterize the spatial radiation of a volume light source in such a way that illuminance can be determined anywhere in space external to the source. It resolves the near field problem and can be used to produce intensity data for simplified far field applications.

My only quibble is the attempt to invoke the concept of helios. This is not necessary and adds nothing to the generality of the concept. The paper suggests that luminance is not general since the definition incorporates the word "surface". However, this only is a convenient way to identify the areal element dA . The author's comment notwithstanding, luminance is a valid concept within a plasma or a dissipative medium. Helios and luminance are equivalent except for a scale factor and the definition of direction along a ray.

Much of the confusion occurs because the definition of helios normally does not explicitly involve an areal element. However, the definition is formulated about flux density, an entity that itself is defined in terms of an areal element dA . This implicit dA in the definition of helios occurs at the point on a ray where helios is determined. The explicit dA in the definition of luminance similarly occurs at the point on a ray where luminance is determined.

Since the practical system of generalized photometry described in this paper is presented in terms of luminance, the initial digression on helios in no way limits the value of the work.

Robert E. Levin, FIES
GTE Products Corp.

Author's Response

To T.L. Ballman:

I agree that a great deal of work needs to be done to validate the practicality of the helios approach. This paper was intended to present its theoretical aspects and to outline what I believe to be a practical approach to near-field photometry. I will not be able to report on its experimental aspects for at least several months.

As a professional engineer, I support Mr. Ballman's KISS philosophy as a design principle. I admit that any implementation of the helios approach will require sophisticated computer hardware and software. However, I maintain that the helios approach is conceptually simple. Its implementation is equally simple in principle -- a video camera mounted on the end of a rotating arm. This apparatus measures luminance values and predicts illuminances. Its performance can be easily validated with a calibrated illuminance meter. Any errors will be immediately and embarrassingly obvious.

To P. Ngai:

Mr. Ngai correctly notes that it is difficult to accurately measure scene luminances using photographic or CCD techniques. However, it is important to remember that the helios approach does not involve imaging the luminaire in any visual sense. In particular, the resolution requirements are minimal in comparison to imaging luminance systems such as the NRC CapCalc. The "point spread" referred to by Mr. Ngai is not a significant problem at low resolutions. Also, the signal-to-noise problems inherent to the CapCalc system can be easily overcome at low resolutions, as I noted in the paper.

I must question Mr. Ngai's estimated computer calculation times of 6 to 24 hours. These figures are reasonable for radiosity-based calculations used to generate an image of a partitioned office space or other complex environment. The hemicube is successively applied to each small patch of each surface, and the solution is obtained iteratively. I contend that near-field calculations need only be performed in the vicinity of the luminaire for the direct illuminance component only. Farther away from the luminaire, far-field calculations will suffice. The computer-generated image can then be calculated using standard ray tracing or radiosity (i.e., radiative transfer) techniques. The near-field calculations should occupy a small portion of the overall computation time.

The simplified approach presented in the paper was intended to provide a more accurate alternative to application distance photometry. As Mr. Ngai correctly notes, it sacrifices some of the advantages of the more general approach. The simplified approach is therefore more suited to lighting design calculations than it is to image generation.

To W.E. Brackett:

Mr. Brackett's comments regarding the need to measure all quadrants of a luminaire are well taken. My planned testing procedures will take this requirement into account.

The hemicube was chosen primarily for its mathematical simplicity. The computer graphics community has developed alternative geometric forms such as the cubic tetrahedron (1,2) and the modified hemi-cube (3) which permit somewhat faster calculations. It is possible to use a hemisphere, but only at the expense of more complicated ray intersection calculations. It is also necessary to maintain a constant element resolution across the surface of the hemicube since the projected view of the luminaire may, depending on the illuminated surface's orientation, occur anywhere on the hemicube surface. The configuration factors are precalculated as per Mr. Brackett's suggestion. The ray positions are similarly determined using 3D transformations of precalculated rays.

The need to know the test distance between the luminaire and the CCD camera is a relevant comment. In particular, the geometric relationship between the imaginary sphere defined by the camera positions and the luminaire must be maintained when later using the near-field data in illuminance calculations. In practice, the geometric center of the

luminaire should be located at the center of the imaginary sphere for both measurement and calculation purposes.

Mr. Brackett raises a number of issues that can only be resolved through experimentation and further software development. I have yet to establish minimum resolution requirements. I have indicated techniques whereby the raw data can be compressed to a manageable size in reasonable amounts of time. I am also fairly confident that the helios approach will yield results comparable in accuracy with those obtained from far-field photometry.

To D. DiLaura:

Mr. DiLaura obviously has the advantage of a better library than I have access to. I credited Dr. Levin in my paper and referenced his 1971 work. I was also aware of the work of Gershun and Yamauti through Moon and Spencer's book "The Photoc Field", but was unable to obtain copies of their papers. I very much appreciate this clarification of the historical record.

The "experimental uncertainties" arose due to the CapCalc experiments having being performed before the theoretical foundations of the helios approach were finalized. The MR-16 lamp had a beam spread of less than 5 degrees, yet the measurements were taken at 2.5 degree intervals. The lamp had a decidedly asymmetrical near-field distribution (less than 150 mm from the lamp's geometric center), yet photometric measurements were only taken in one vertical plane. The small size of the lamp made near-field illuminance measurements difficult -- the photosensor was larger than the spacing increment. Finally, the test lamp was misplaced before surface illuminance measurements could be performed. These difficulties in no manner relate to the practicality of the helios approach.

The theoretical results are useful in that they demonstrate the capabilities of the approach in the absence of any measurement uncertainties. A perfectly diffuse emitting disk was used mostly because its analytic radiative transfer function is so well known; the same approach can be applied to an area or volume source with any degree of geometric complexity as long as its radiative transfer function can be solved analytically or numerically.

It must be remembered that the helios approach is independent of the light source geometry. A surface luminance resolution of 12 mm is required only if the illuminated surface is 60 mm from the luminaire (if the Five-Times Rule is adopted). The luminance distribution within the field of view of each pixel is integrated with no significant loss of illuminance information. A resolution of 64x64 pixels allows the illuminance calculation point to be as close as 100 mm to a 1.2 meter diameter light source. The distance of the

compound luminance meter from the light source is irrelevant as long as the source fills its field of view.

I have not yet addressed the minimum angular resolution required for typical luminaires. While relatively simple to solve theoretically, this issue must consider the geometric and photometric complexity of a wide range of commercially available products. On the other hand, there is no reason why the measurement process cannot be made adaptive by taking measurements at (say) 0.5 degree intervals and retaining only those which are significantly different. The same technique is used in the ISO/CCITT MPEG image compression technique.

The luminance resolution of 256 steps is determined primarily by the capabilities of existing CCD technology. While research-grade CCD cameras with luminance resolutions of over 65,000 steps are available, they are prohibitively expensive. The use of a separate calibrated illuminance meter permits the use of the camera's automatic gain control (AGC) and electronic shutter circuitry to cover a much wider luminance range (although not within the same field of view). I am currently experimenting with a Cohu Series 4910 CCD camera, whose AGC circuitry can be adjusted to respond to peaks (highlights) in the field of view or the overall average light level. Mr. DiLaura's comments regarding luminance resolution are valid; they can only be resolved by experimentation, which is now in progress.

To Robert E. Levin:

Dr. Levin apparently misread my comments on luminance and helios. Referencing Dr. Levin's own paper "Luminance - A Tutorial Paper", I noted that helios is no more than luminance defined for a (real or imaginary) surface. I also agree that luminance is a valid concept within a plasma or dissipative medium. I invoked the concept (or, to use a more modern term, paradigm) of helios not only to emphasize the point that the ANSI/IES definition of luminance applies to imaginary as well as real surfaces, but to honor the work of Drs. Parry Moon and Domina Spencer. As Dr. Moon noted in his paper "A System of Photometric Concepts" (4), it is not necessary to adopt the helios nomenclature as long as we fully recognize and appreciate the concepts involved.

References

1. Arvo, J. [1991]. *Graphic Gems II*, Academic Press, San Diego, CA.
2. Beran-Koehn, J.C. and M.J. Pavicic [1991]. "A Cubic Tetrahedral Adaptation of the Hemi-Cube Algorithm", in Arvo, 299 - 302.

3. Recker, R.J., D.W. George and D.P. Greenberg [1990]. "Acceleration Techniques for Progressive Refinement Radiosity", Computer Graphics 24:2 (1990 Symposium on Interactive 3D Graphics), 59 - 66.
4. Moon, P. [1942]. "A System of Photometric Concepts", J. Optical Society of America 32 (June), 348 - 362.

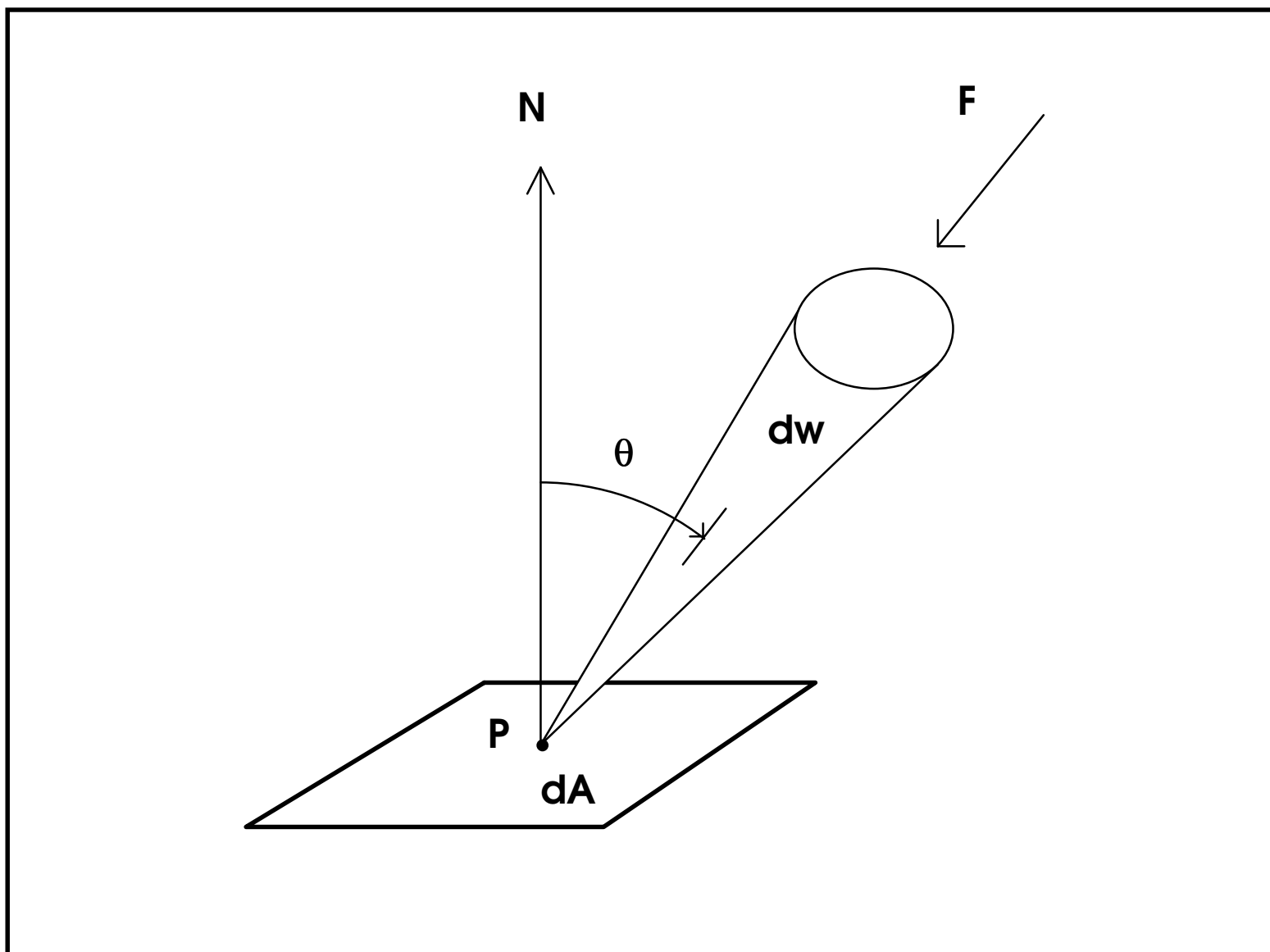


Figure 1

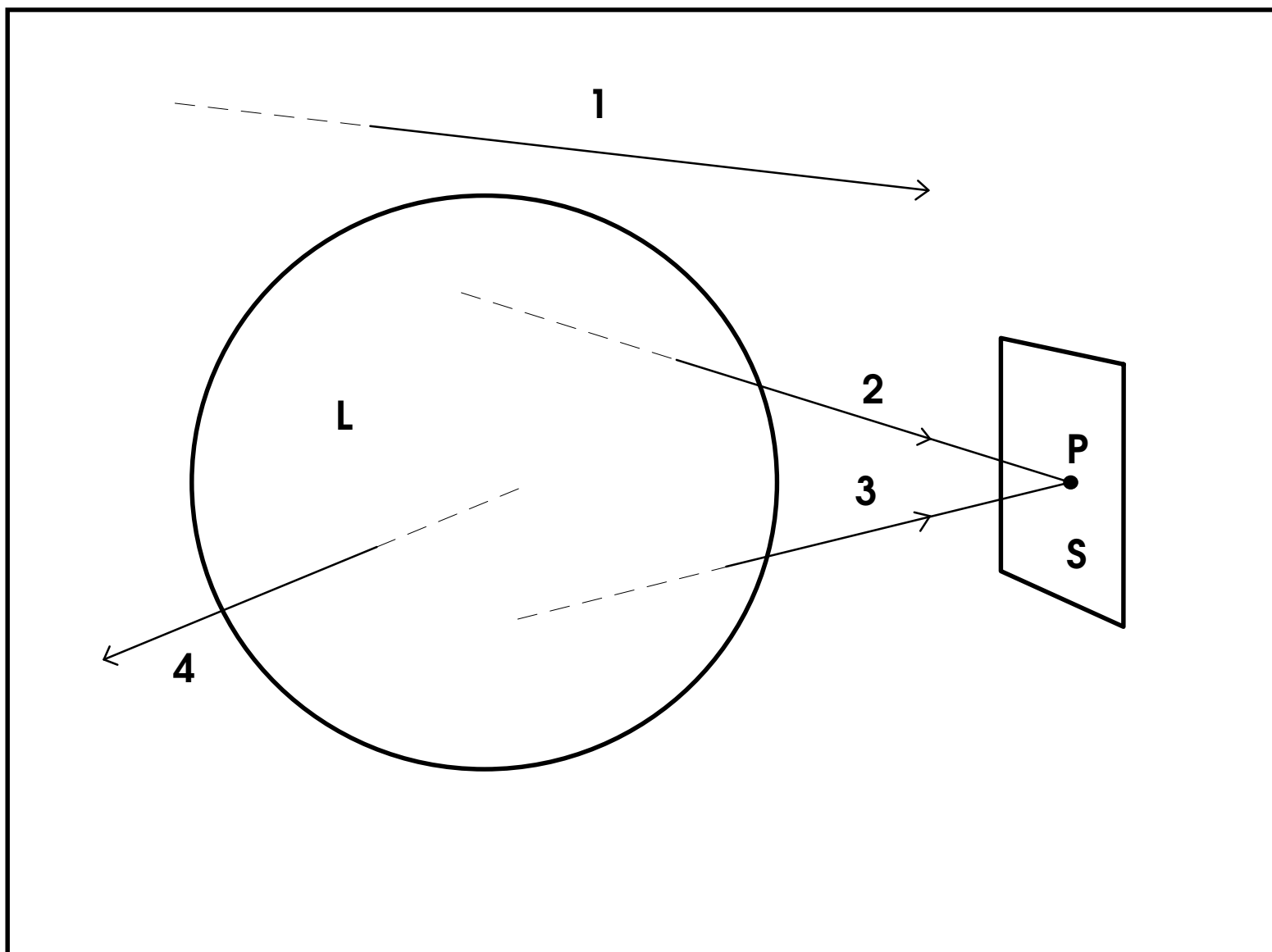


Figure 2

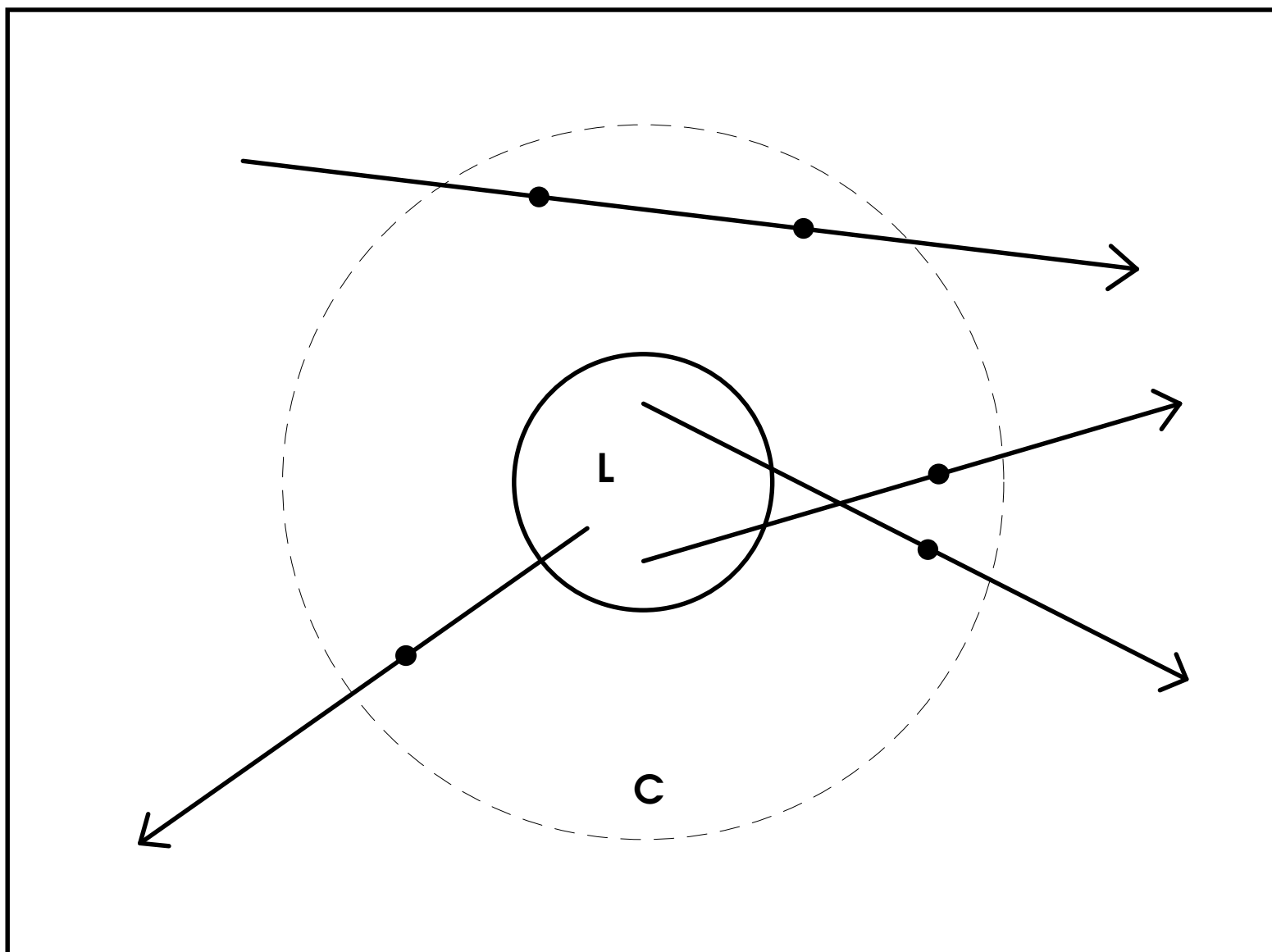


Figure 3

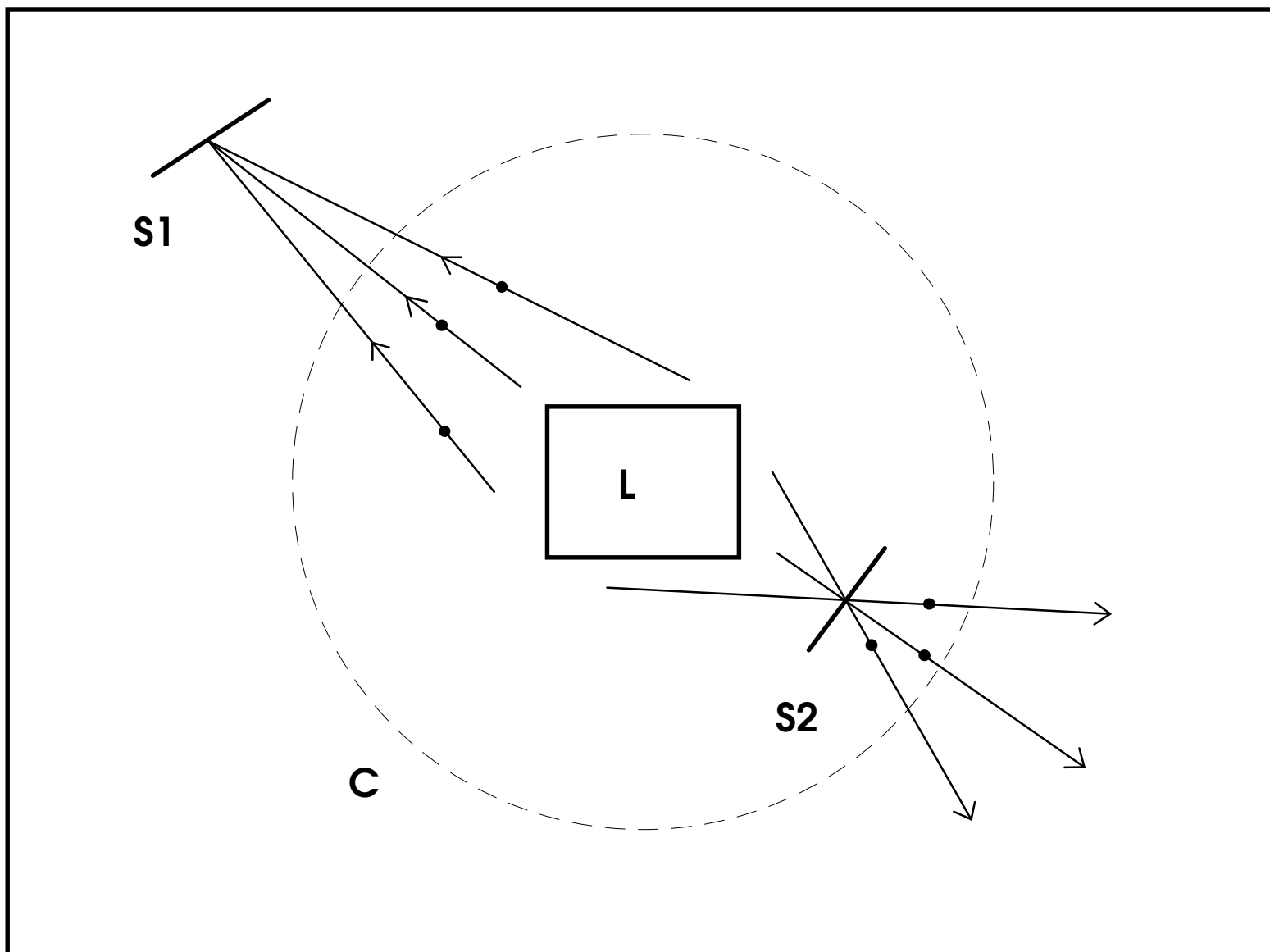


Figure 4

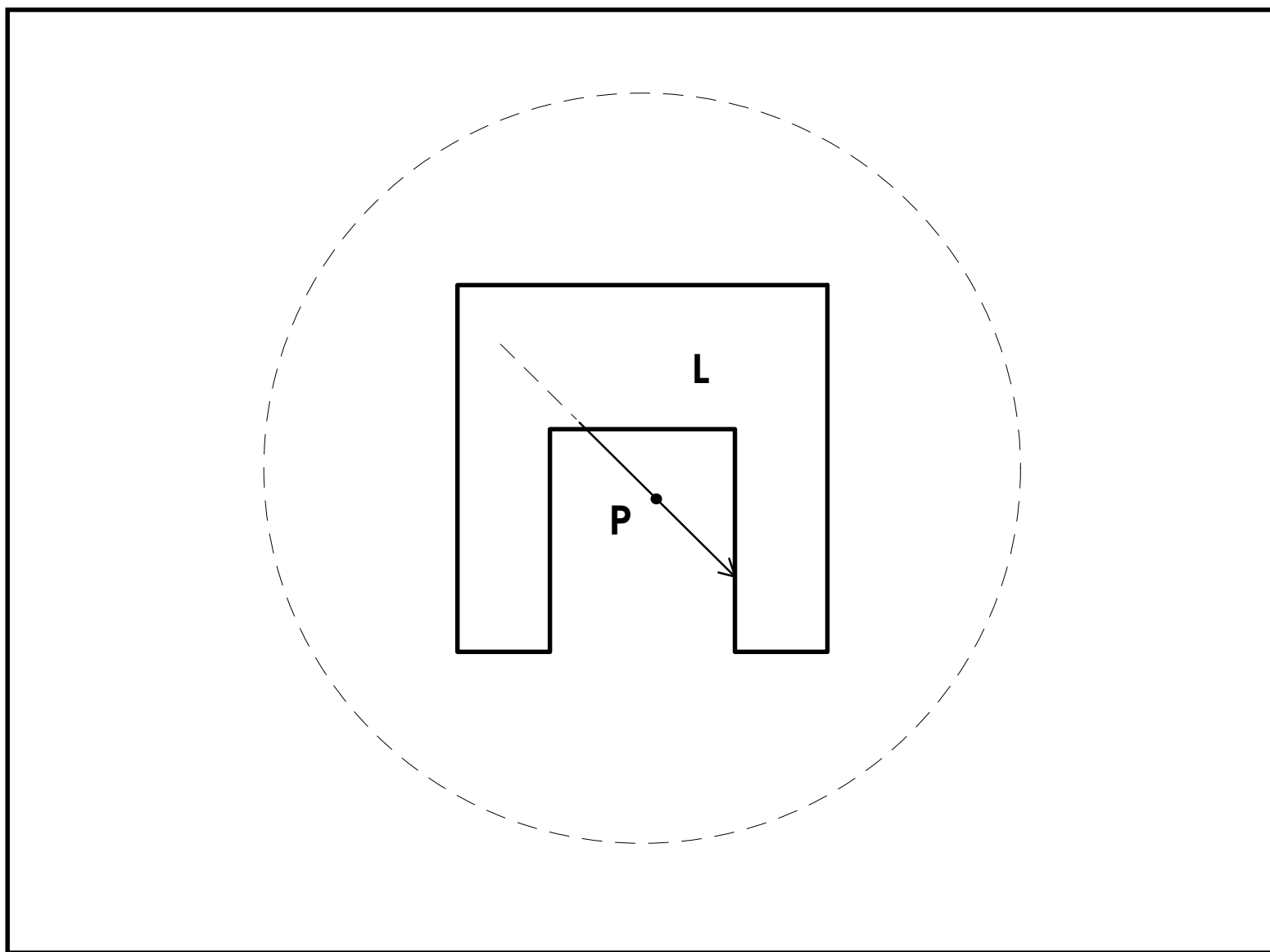


Figure 5

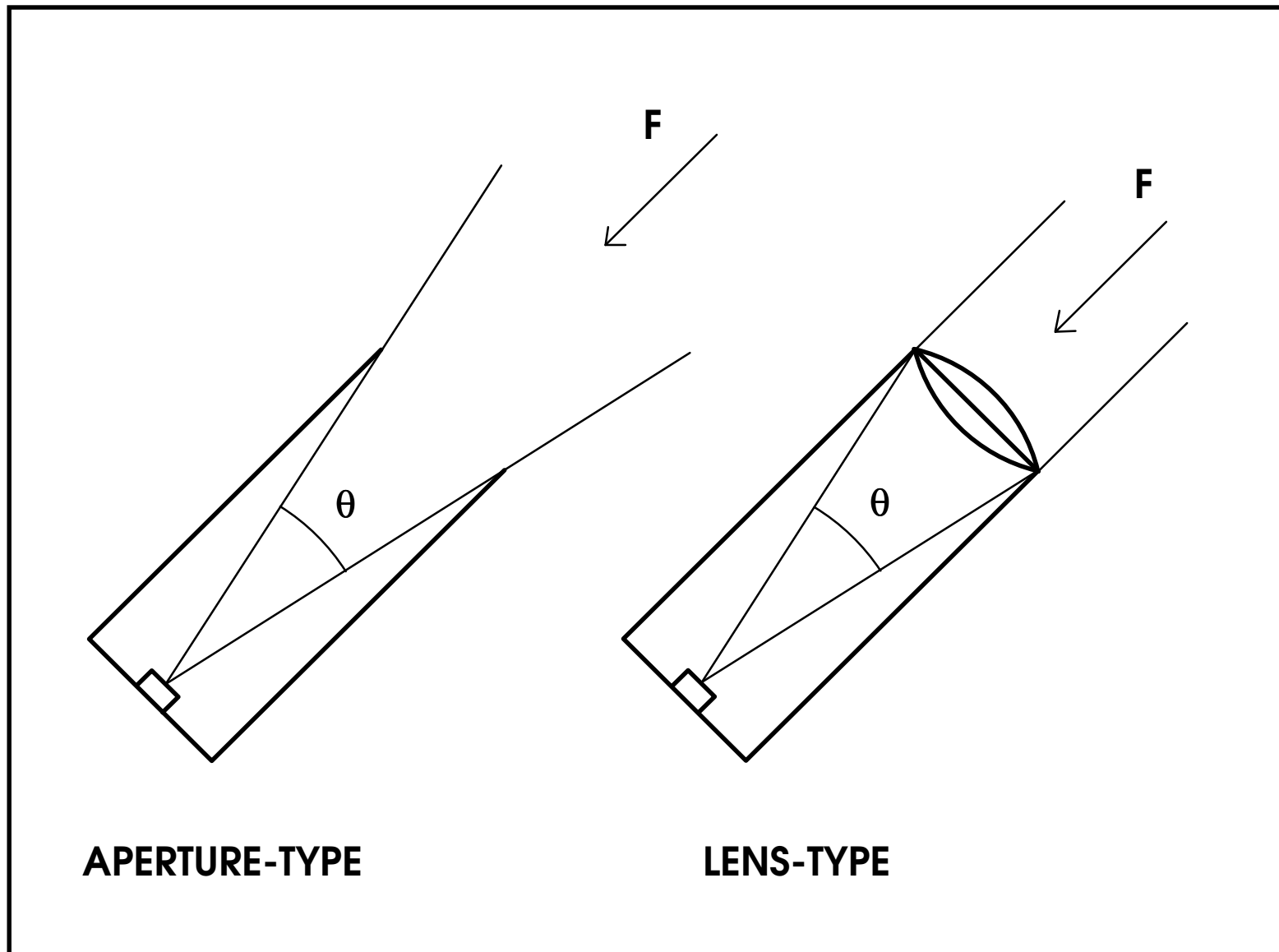


Figure 6

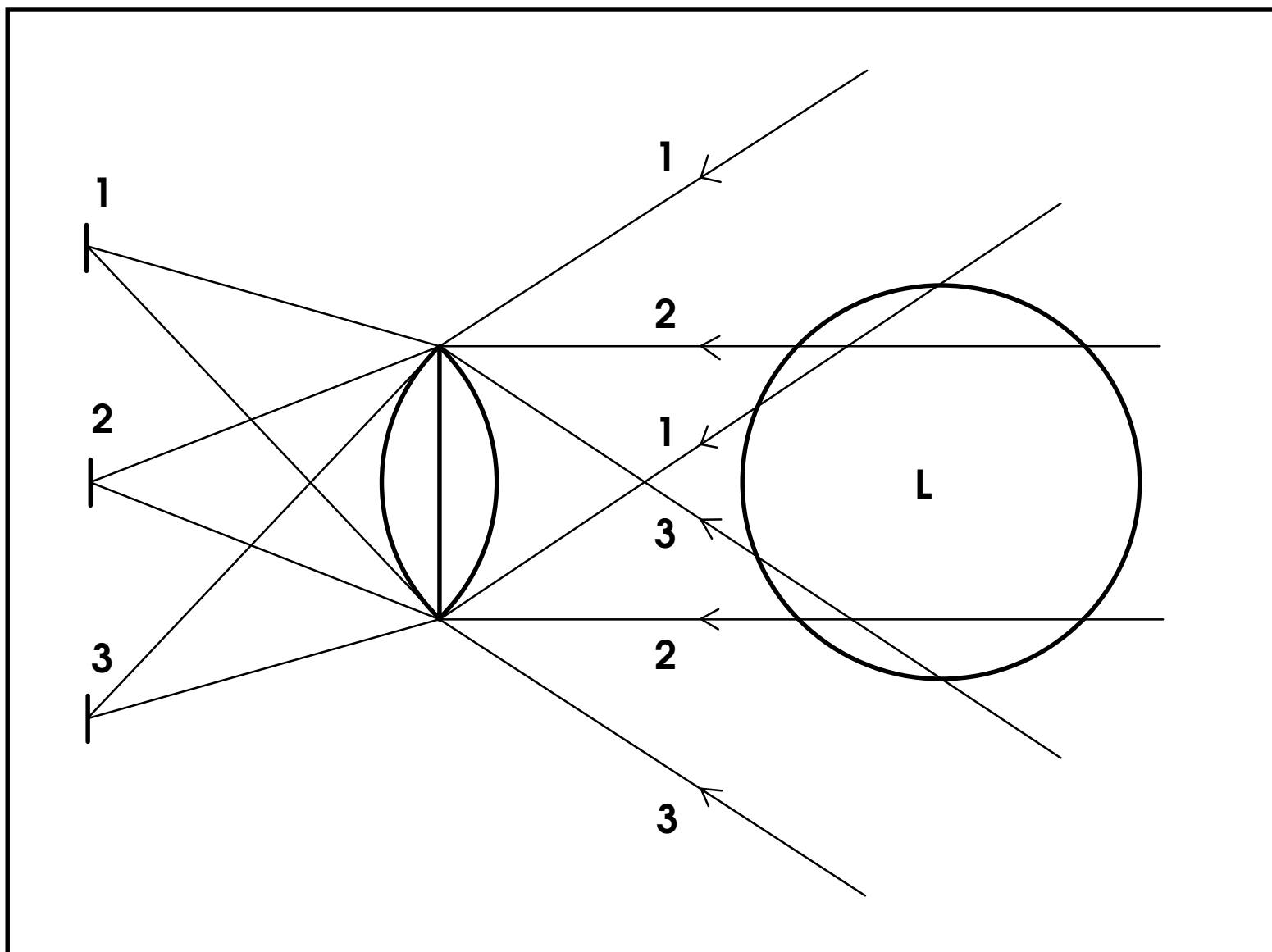


Figure 7

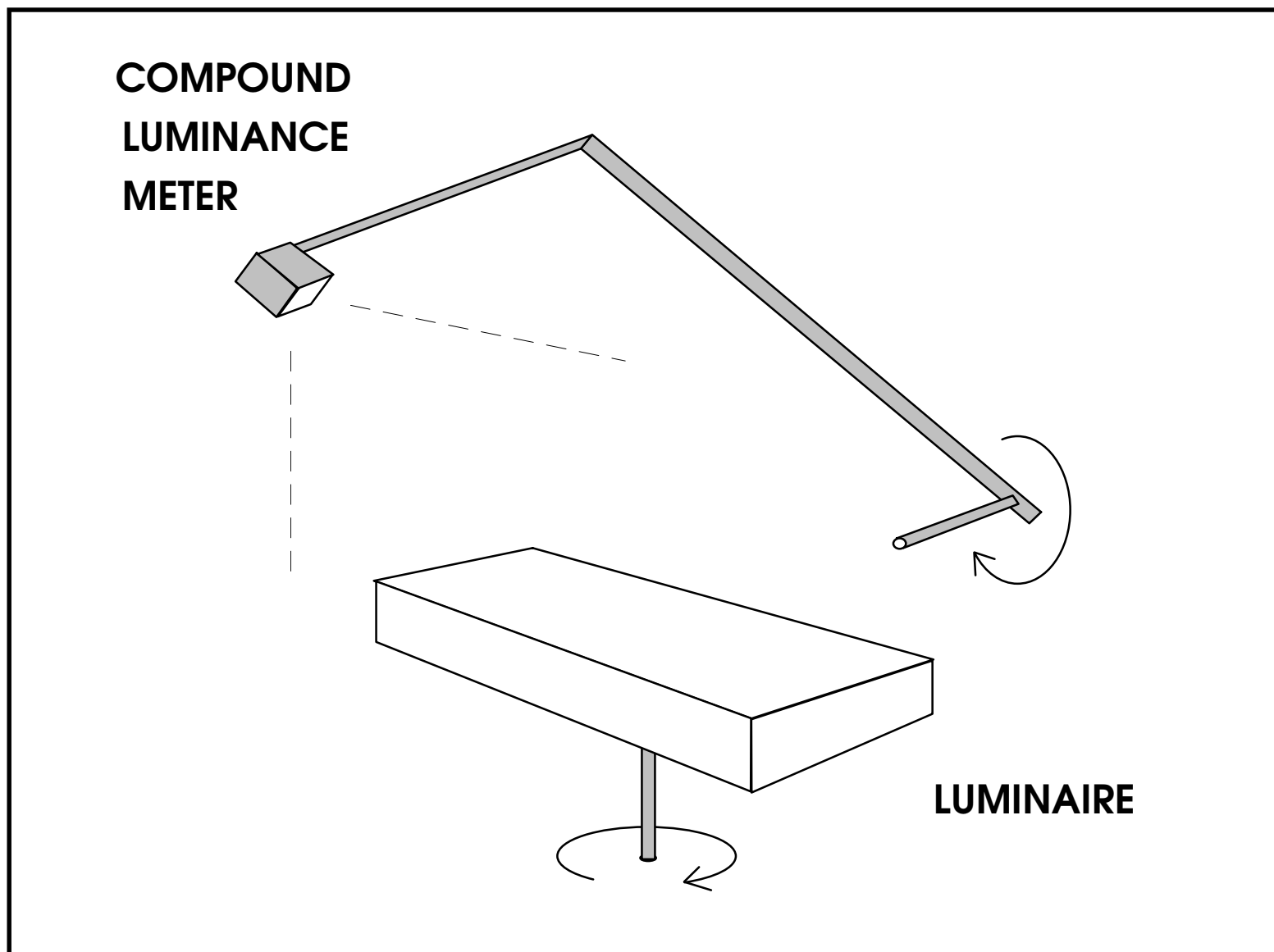


Figure 8

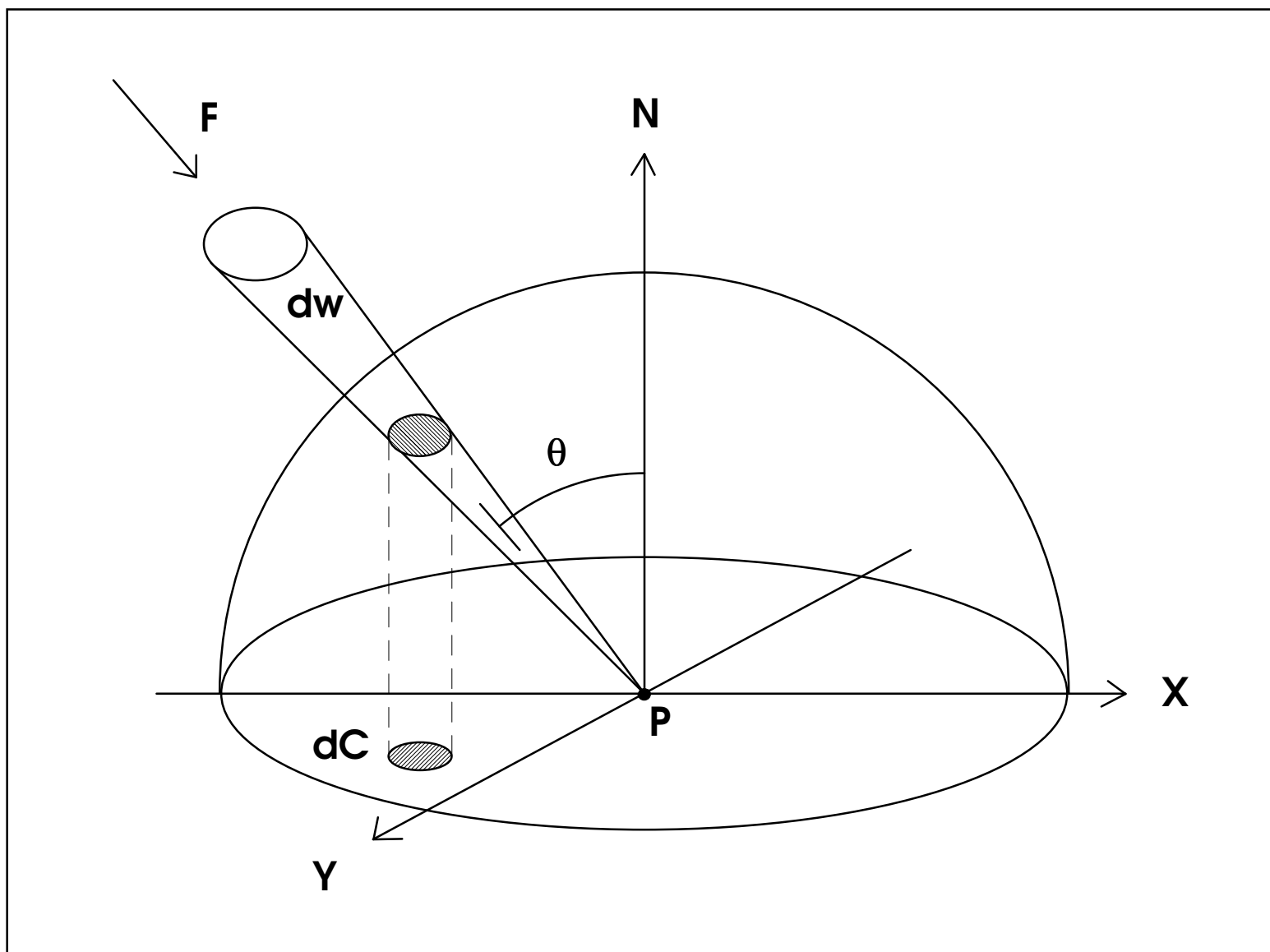


Figure 9

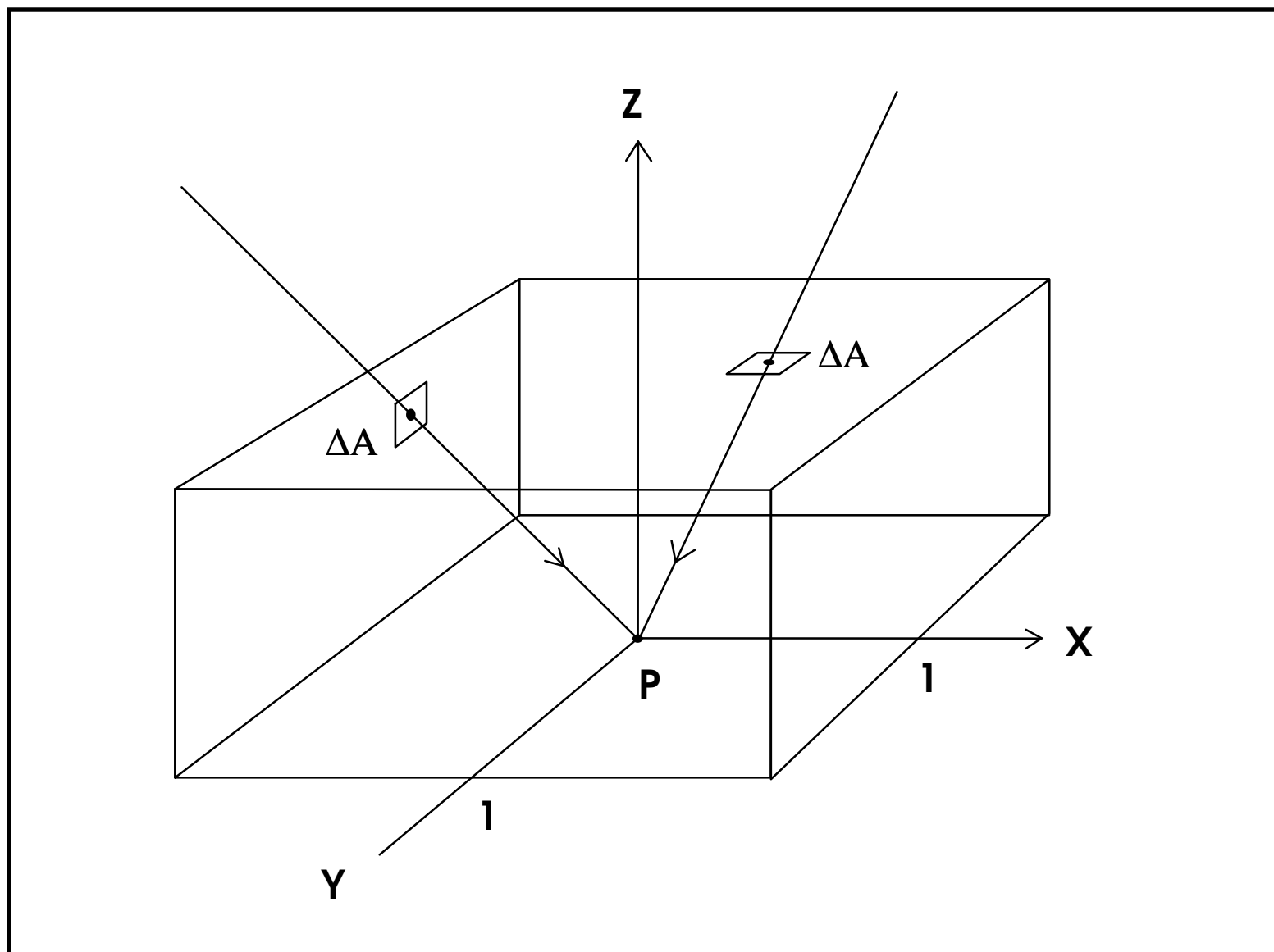


Figure 10

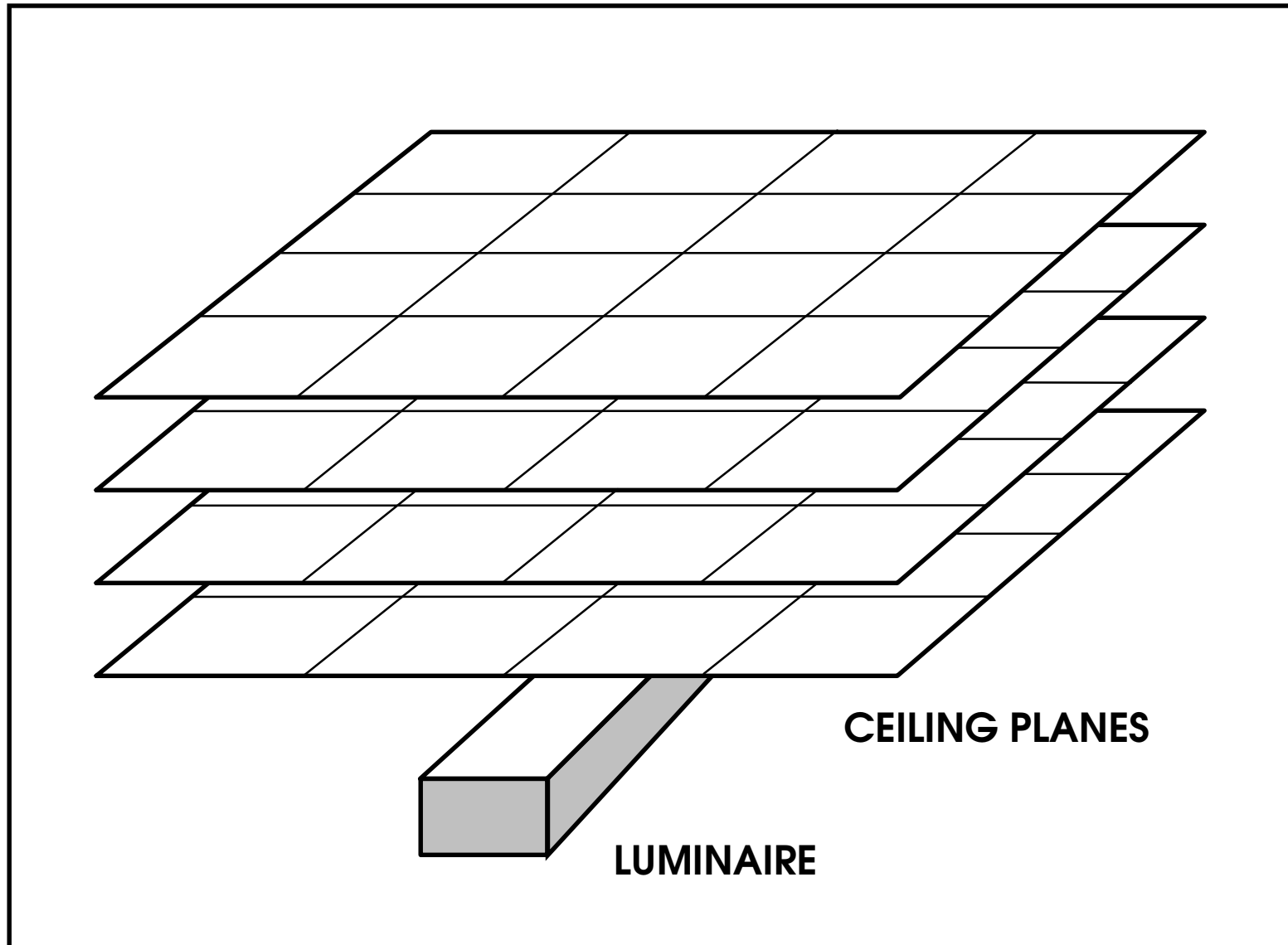


Figure 11

LEDALITE NEAR-FIELD PHOTOMETRIC ANALYSIS

FILE INFORMATION

Photometric File: SYN_C006.NFP
Report Number: LAP-0000
Description: Synthetic Luminous Disk

VIEWING POSITION

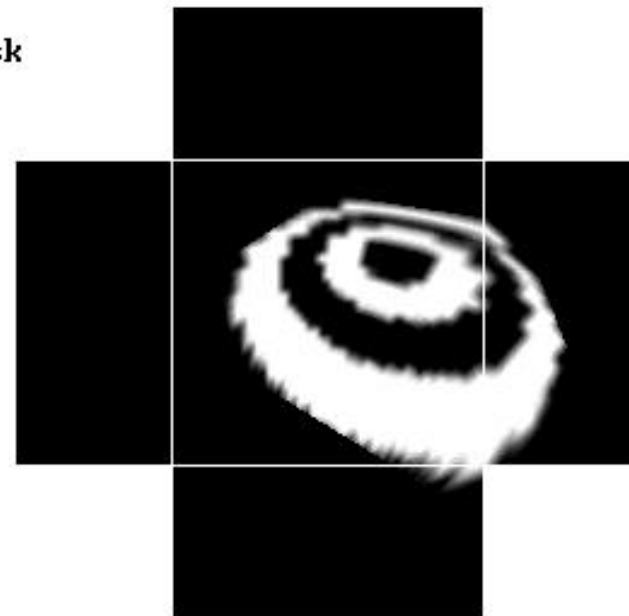
Horizontal Angle: 0.00 degrees
Vertical Angle: 120.96 degrees
View Distance: 0.58 meters

HEMICUBE PARAMETERS

Plot Type: Pseudocolor
Resolution: 64 x 64
Horizontal Angle: 206.57 degrees
Vertical Angle: 36.70 degrees

ILLUMINANCE PREDICTION

Illuminance: 26.4403 lux
Maximum Luminance: 100.0000 cd/m2



SYNTHETIC IMAGE

Press ALT-G for grey scale

LUMINANCE



0

25

50

75

100

Figure 12A

LEDALITE NEAR-FIELD PHOTOMETRIC ANALYSIS

FILE INFORMATION

Photometric File: SYN_C004.NFP
Report Number: LAP-0000
Description: Synthetic Luminous Disk

VIEWING POSITION

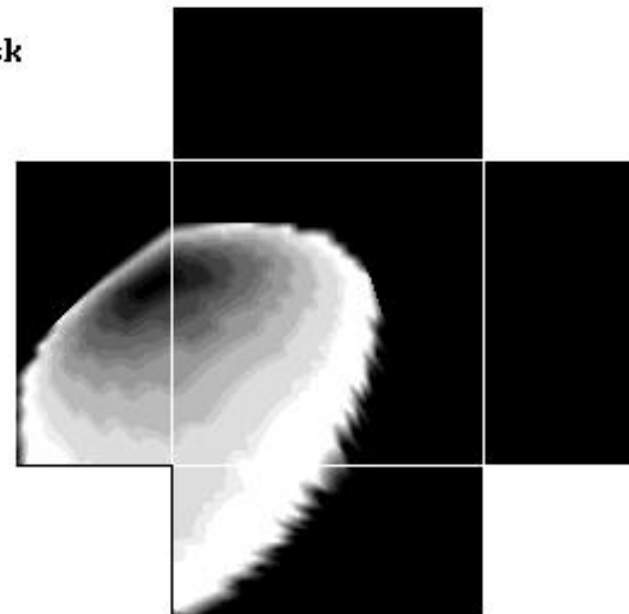
Horizontal Angle: 53.13 degrees
Vertical Angle: 111.80 degrees
View Distance: 0.54 meters

HEMICUBE PARAMETERS

Plot Type: Grey scale
Resolution: 64 x 64
Horizontal Angle: 180.00 degrees
Vertical Angle: 45.00 degrees

ILLUMINANCE PREDICTION

Illuminance: 31.8576 lux
Maximum Luminance: 100.0000 cd/m²



SYNTHETIC IMAGE

Press ALT-C for pseudocolor

LUMINANCE



0

25

50

75

100

Figure 12B

LEDALITE NEAR-FIELD PHOTOMETRIC ANALYSIS

FILE INFORMATION

Photometric File: SYN_C001.NFP
Report Number: LAP-0000
Description: Synthetic Luminous Disk

VIEWING POSITION

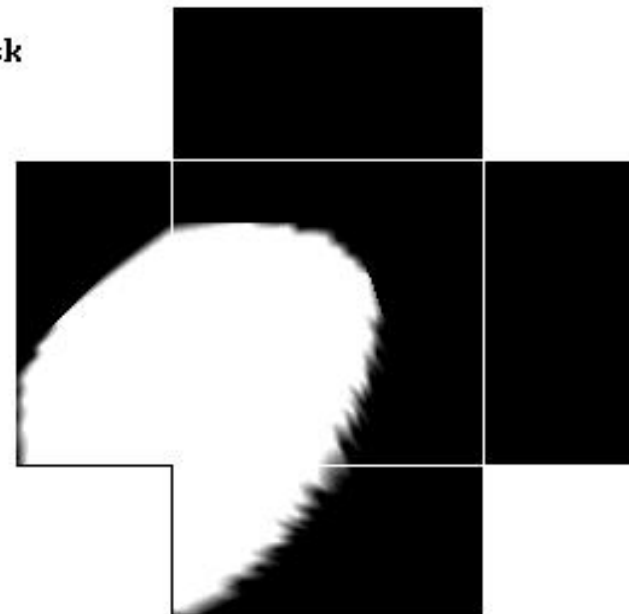
Horizontal Angle: 53.13 degrees
Vertical Angle: 111.80 degrees
View Distance: 0.54 meters

HEMICUBE PARAMETERS

Plot Type: Pseudocolor
Resolution: 64 x 64
Horizontal Angle: 180.00 degrees
Vertical Angle: 45.00 degrees

ILLUMINANCE PREDICTION

Illuminance: 42.1657 lux
Maximum Luminance: 100.0000 cd/m²



SYNTHETIC IMAGE

Press ALT-G for grey scale

LUMINANCE

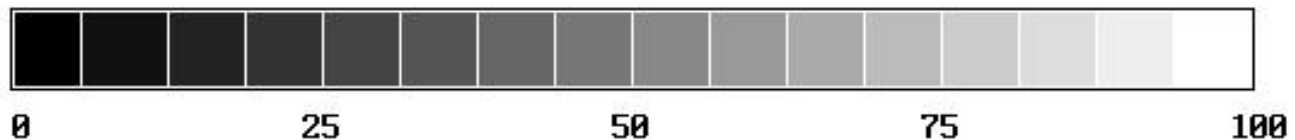


Figure 12C

